

A Modal Description of Dynamic Wake Meandering

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Horizontal scans from nacelle-mounted lidar systems provide time series of wake measurements during three different atmospheric inflow conditions, from which we describe the coherent turbulent structures that contribute to wake meandering through proper orthogonal decomposition. Subsets of modes are used to make low-order flow field reconstructions in a combinatorial sense, yielding more than 16,000 estimates of meandering for each of the inflow cases. A regression test using the range of reconstructed flow statistics identifies the modes that contribute most to the accurate description of wake meandering. Spectra defined from each mode coefficient time series highlight the dominant Strouhal number associated with each coherent turbulent structure, and suggest that the lowest ranking modes do not necessarily contribute most to the accurate representation of wake meandering. Instead, some modes appear to have no influence on meandering dynamics, and still others consistently work against the accurate description of wake meandering in reduced-order models. No consistent relationship is revealed between characteristic frequencies for each mode and for either the inflow or the lidar measurements, suggesting that a more complex relationship between wake and inflow turbulence may be needed to accurately describe wake meandering.

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I. INTRODUCTION

Wake meandering, characterized by large-scale, quasiperiodic, low-frequency oscillations of the entire wind turbine wake, plays a crucial role in wind turbine loads, controller set point uncertainty, and power quality. Meandering significantly impacts power production and fatigue loading of downstream turbines in wind farms.^{1–3} Despite its importance, accurately representing wake meandering in numerical models remains challenging at any level of fidelity, making any additional characterization of the phenomenon valuable for improving wind farm performance and reliability.

Two prevailing hypotheses attempt to explain the origin of wake meandering. The externally driven hypothesis considers the wake’s momentum deficit as a passive tracer advected by large-scale atmospheric turbulence.¹ This theory suggests that only turbulent structures larger than the rotor diameter significantly contribute to wake meandering.^{1,4–8}

Alternatively, the internally driven hypothesis proposes that dynamic interactions between turbulent structures within the wake itself, such as tip vortices and hub/root vortices, generate pressure fluctuations substantial enough to displace the wake.^{9–14} Also within the internally-driven set of hypotheses is that wake meandering may result from a fundamental shear instability that can be excited by multiple mechanisms.^{15,16}

Our ability to measure wind turbine wakes in the field has improved tremendously over the past decade. Ten years ago, some of the first dynamic wake measurements were collected by modifying a prototype lidar originally designed for vertical conical scanning.^{17,18} These early measurements, despite their limitations, were used to validate basic assumptions of the dynamic wake meandering (DWM) model.^{1,7} Today, lidar technology allows us to observe wake dynamics for wind turbines of any size, with the capability to scan in any direction spanwise and vertically, reach downstream distances on the order of kilometers, and capture wake meandering, the downstream trajectory of wake centers, expansion, and large-scale turbulent structures.² Nacelle-mounted lidars have emerged as a powerful tool for real-time characterization of inflow and wakes of utility-scale wind turbines.¹⁹ Doppler lidars offer unique capabilities for wind energy research, including real-time measurements for turbine and farm control, long-term performance assessment, high-resolution data assimilation for numerical simulations, and detailed wake characterization.^{18,20–26} provide better characterization of the wind resource at relevant heights and allow effortless steering onto the wind

direction for yaw-controlled turbines.

Several recent studies leverage modern lidar technology to characterize turbulence in the wake. Depending on the instrument deployed, lidars typically measure in vertical planes at fixed downstream distances,^{25,27,28} or in horizontal planes perpendicular to the rotor at hub height.^{29,30} These recent, lidar-based studies have primarily looked at the temporal evolution of turbulence by identifying the wake within each scan and tracking its centerline position over time. The centerline is then used to quantify wake meandering for the sake of wake characterization²⁷ or model validation.²⁵

To investigate under what conditions we can expect the internally-driven hypothesis to be the dominant mechanism driving wake meandering, we employ Proper Orthogonal Decomposition (POD). This method is well-suited for identifying coherent turbulence structures in flow fields³¹ and has been applied to wind fields for over fifteen years.^{32–36} POD seeks the variance-maximizing structures that describe the turbulent kinetic energy in the flow, making it an ideal tool for quantifying potential meandering-inducing structures in the wake. Our approach involves developing a reduced-order model (ROM) using the POD-derived structures. If these structures are shown to contribute significantly to wake meandering, the ROM should reproduce the meandering behavior seen in the measurement data to a high level of fidelity. Conversely, a ROM developed without these structures should fail to accurately reproduce wake meandering.

Through the use of POD, we will estimate the meandering length scales and frequencies in our dataset and relate them to DWM modeling assumptions. The body of past experimental and numerical work typically characterizes the meandering frequency (f_m) in the context of a Strouhal number, $St = f_m D / U_{\text{hub}}$ defined in terms of the wind turbine rotor diameter (D) and hub-height inflow velocity (U_{hub}). Meandering frequencies are typically reported in the range of $0.1 \leq St \leq 0.35$, both in laboratory and field observations.^{2,37,38} If the internally-driven hypothesis for wake meandering is correct, the modes that contribute most to meandering should have characteristic frequencies higher than the inflow Strouhal number. Conversely, cases where the externally driven hypothesis is correct should favor structures with frequencies below the inflow Strouhal number.

DWM models^{1,8} consider only the inflow forcing, assuming the momentum deficit and wake center to advect as passive tracers in a background flow. Typically, this effect is considered by applying a low-pass filter to the turbulent inflow such that only structures that

exceed a certain minimum length scale influence wake motion. This cut-off length scale is often set to twice the observed instantaneous wake diameter, or simply as twice the rotor diameter ($2D$) in most implementations.^{39,40} By combining advanced measurement techniques with sophisticated data analysis methods, we seek to develop an understanding of wake meandering across a wide range of atmospheric conditions. We relate and its impact on wind farm performance, ultimately contributing to the optimization of wind energy production and turbine longevity.

II. METHODS

A. Wake meandering

Wake meandering is quantified in this work through the dynamics of the transverse coordinate of the wake center (μ), which we estimate by fitting a Gaussian profile to the momentum deficit derived from the lidar scans. The velocity deficit is defined as,

$$\tilde{u}(x, y, t) = 1 - \frac{u(x, y, t)}{U_{\text{hub}}(t)} \quad (1)$$

In Eq. (1), u represents the streamwise velocity measurement, estimated from the cosine of the line-of-sight velocity observed by the lidar. Because our data are collected in approximately horizontal planes at hub height, we only consider lateral movements of the wake center along the y direction at a fixed height above ground $z = z_{\text{hub}}$. The wake center is detected through a least-squares fit of a Gaussian function to the observed U by the lidar at each downstream location and for each time.

$$\tilde{u}(x, y, t) = A \exp \frac{1}{2} \left[\frac{y - \mu(x, t)}{\sigma(x, t)} \right]^2 + C \quad (2)$$

In Eq. (2), the peak momentum deficit is denoted as A , the standard deviation corresponding to wake width is σ , and the lateral wake center is μ . An offset term, C , is included in the function, although it is typically a relatively small value, $-1 \lesssim C \lesssim 1$.

B. Snapshot Proper Orthogonal Decomposition

Proper Orthogonal Decomposition (POD), also known as Principal Component Analysis (PCA) or Karhunen-Loève decomposition, has been widely used in fluid dynamics for identifying coherent structures and reducing the dimensionality of complex flow fields³¹. However,

its application to data in polar coordinates, as is more natural for scanning lidar observations, presents unique challenges that necessitate a careful reconsideration of the method's foundational elements.

Snapshot POD begins with a sequence of lidar observations $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_m]$ where each snapshot is a realization in the state space $\mathbf{x}_k \in \mathbb{R}^n$ at time t_k . POD seeks an orthonormal basis that optimally represents the data in terms of captured variance. When dealing with data in polar coordinates, the standard Euclidean inner product implicit in this formulation becomes inappropriate to adequately represent the distribution of turbulent kinetic energy in the domain. The varying cell sizes in a polar grid lead to an inherent bias, overemphasizing contributions from larger radii. To address this, we modify the standard POD algorithm to incorporate a physically appropriate inner product for polar coordinates. Considering two snapshots $\mathbf{x}(r, \theta)$ and $\mathbf{y}(r, \theta)$, each of which represents the streamwise velocity (system state) vector in polar coordinates, we define a weighted inner product as,

$$\langle \mathbf{x}, \mathbf{y} \rangle_W = \int_0^{2\pi} \int_0^R \mathbf{x}(r, \theta) \cdot \mathbf{y}(r, \theta) r dr d\theta \quad (3)$$

For discrete data on a polar grid with spacing Δr and $\Delta \theta$ in the radial and azimuthal directions, respectively, Eq. (3) becomes,

$$\langle \mathbf{x}, \mathbf{y} \rangle_W = \sum_{i=1}^{N_r} \sum_{j=1}^{N_\theta} \mathbf{x}(r_i, \theta_j) \cdot \mathbf{y}(r_i, \theta_j) r_i \Delta r \Delta \theta \quad (4)$$

The inner product in Eq. (4) can be represented by a diagonal weight matrix $\mathbf{W}$:

$$\mathbf{W} = \text{diag}(r_1, \dots, r_1, r_2, \dots, r_2, r_3, \dots, r_{N_r}, \dots, r_{N_r}) \Delta r \Delta \theta \quad (5)$$

where each radial coordinate, r_i , is repeated N_θ times.

The new inner product is incorporated into the POD algorithm by modifying correlation matrix to account for the weights described by Eq. (5) as,

$$\mathbf{C} = \frac{1}{m} \mathbf{X}' \mathbf{W} \mathbf{X} \quad (6)$$

The POD modes are then obtained by solving the eigenvalue problem in the standard fashion,

$$\mathbf{C} \mathbf{v}_i = \lambda_i \mathbf{v}_i \quad (7)$$

where λ_i are the eigenvalues and $\mathbf{v}_i$ are the corresponding eigenvectors. The POD modes ϕ_i are then reconstructed as:

$$\phi_i = \frac{1}{\sqrt{\lambda_i}} \mathbf{X} \mathbf{v}_i \quad (8)$$

This formulation ensures that the POD modes accurately reflect the true spatial scales and turbulent kinetic energy of the flow structures in polar coordinates, without bias towards structures at larger radii.

This adaptation of the standard snapshot POD algorithm maintains the core principles of POD while adapting it to the geometric constraints of polar coordinate systems, aligning with the theoretical perspective advocated by Holmes⁴¹, that the choice of inner product in modal decompositions should reflect the physics of the problem at hand. By incorporating an inner product motivated by the polar description of the state vector, we extend the applicability of POD to the natural coordinate system of scanning Doppler lidars, and open the method to a wider class of atmospheric science and wind energy problems.

C. Reconstruction

The approach to velocity field reconstruction used in this work relies on the nature of the POD to organize input system dynamics into coherent structures. We develop a combinatorial approach to reduced order modeling to identify the turbulent structure (POD modes, ϕ_i) that contribute most to wake meandering. Each combination of modes was determined by selecting k of the first N_r modes, where $k \in [3, \dots, 15]$ and $N_r = 15$ was determined by requiring at least 90% of the TKE represented in the observational basis for each case. Flow fields reconstructed with each unique combination of modes are then used to estimate wake meandering identically as for the lidar scans, as described in Section II A.

In each of the ROMs, the zeroth mode, ϕ_0 , representing the mean flow, is also included. The total number of combinations for each dataset was $K = \sum_{k=3}^{14} \binom{N_r}{k} = 16,278$. Fig. 1 shows the number of combinations tested for each value of k .

For each ROM, velocity fields are reconstruction from the truncated basis, summing over selected mode indices, rather than a sequential set up to a maximum. For each combination $I = \binom{N}{k}$, the reconstructed velocity field is,

$$\hat{u}(\mathbf{x}, t) = \sum_{i \in I} a_i(t) \Phi^{(i)}(\mathbf{x}) \quad (9)$$

where the index i may take only the values included in a particular combination of modes and the hat notation indicates a low-dimensional representation of the velocity field. In Eq. (9), the velocity field and modal basis have been denoted as scalars, as only the streamwise

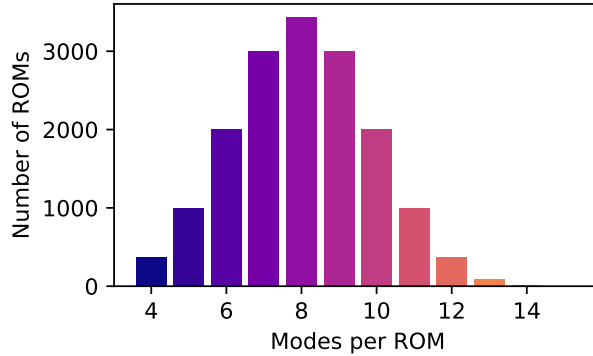


FIG. 1: Number of mode combinations used to reconstruct the flow field for a given maximum number of modes considered (k). For all cases, only modes 1 through 15 are considered ($N_r = 14$).

component of velocity is considered.

III. DATA

This work makes use of nacelle-mounted scanning lidar data and atmospheric inflow measurements from two recent DOE-funded field campaigns. The American WAKE Experiment (AWAKEN⁴²) is a multi-institutional field campaign designed to gather critical data on wind farm-atmosphere interactions, which contribute to uncertainty in wind plant performance models. Atmospheric conditions were characterized through a combination of metrics derived from a surface flux station⁴³ and a meteorological tower⁷ located at site A1. The surface flux station features a Gill R3-50 sonic anemometer mounted on a 4-meter-high tripod, recording data continuously at a 20 Hz rate. Data acquisition used software originally developed by Argonne National Laboratory for the ARM ECOR system in 2003, which was adapted for the AWAKEN project to work with Gill Sonic R3 series anemometers.⁴⁴ Sensible heat flux and momentum flux were processed into data at 30 minute resolution, assuming a constant relative humidity of 50% due to the absence of real-time humidity measurements. Hub-height wind speed and wind direction were recorded with a Thies 3D Ultrasonic anemometer. Turbulence intensity was estimated as a 10-min rolling calculation of the standard deviation of wind speed divided by mean wind speed in the same period. Wind directions were limited to a sector ranging from 170° to 200°, corresponding to the peak of the wind rose shown in Fig. 2. This also limited times when the wake from a

neighboring turbine were visible in the data.

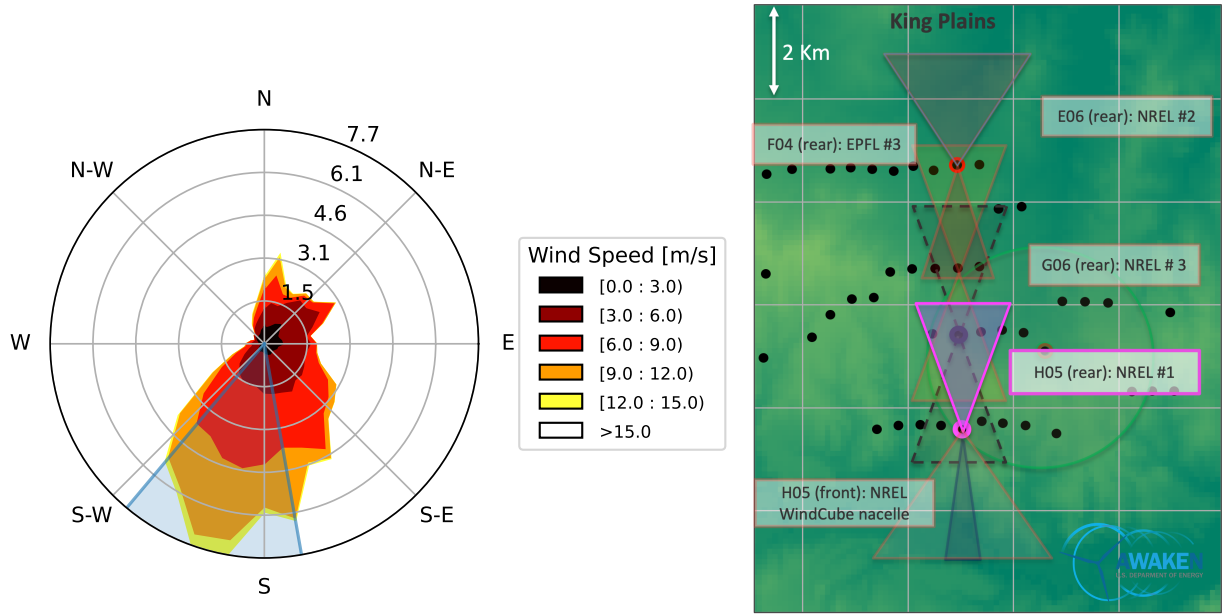


FIG. 2: Wind rose showing selected wind direction sector in blue (left) and high-level schematic (right) of the AWAKEN project.

The Rotor Aerodynamics, Aeroelastics, and Wake (RAAW) project is a field campaign designed to study the performance of modern flexible rotor wind turbines. Atmospheric inflow conditions for data cases in the RAAW project are derived from a 183.5-meter guyed meteorological tower, instrumented between the rotor bottom and top heights with 5 cup anemometers, 3 ultrasonic anemometers, 3 vanes, 3 barometric and temperature sensors, and one humidity sensor. Ultrasonic anemometers reported data at 20 Hz, and all other instruments at 1 Hz. Near-surface measurements of momentum and heat flux were provided by a surface met station, deployed approximately 10 meters southwest of the meteorological tower, that hosted an ultrasonic anemometer measuring at 2.5 m above ground and a probe for atmospheric pressure, temperature and humidity 2 m above ground. Similarly to the tall tower sensors, the anemometer reported data at 20 Hz and the other sensors at 1 Hz. **Figure XXX** shows the experimental arrangement of the RAAW project, including the met tower, surface met station, and turbine. **NEED A RAAW CITATION, when will the report be done?**

Both the RAAW and the AWAKEN project feature GE 2.8 MW turbines with 127 m rotor diameters. The RAAW project made use of a prototype R&D turbine located in

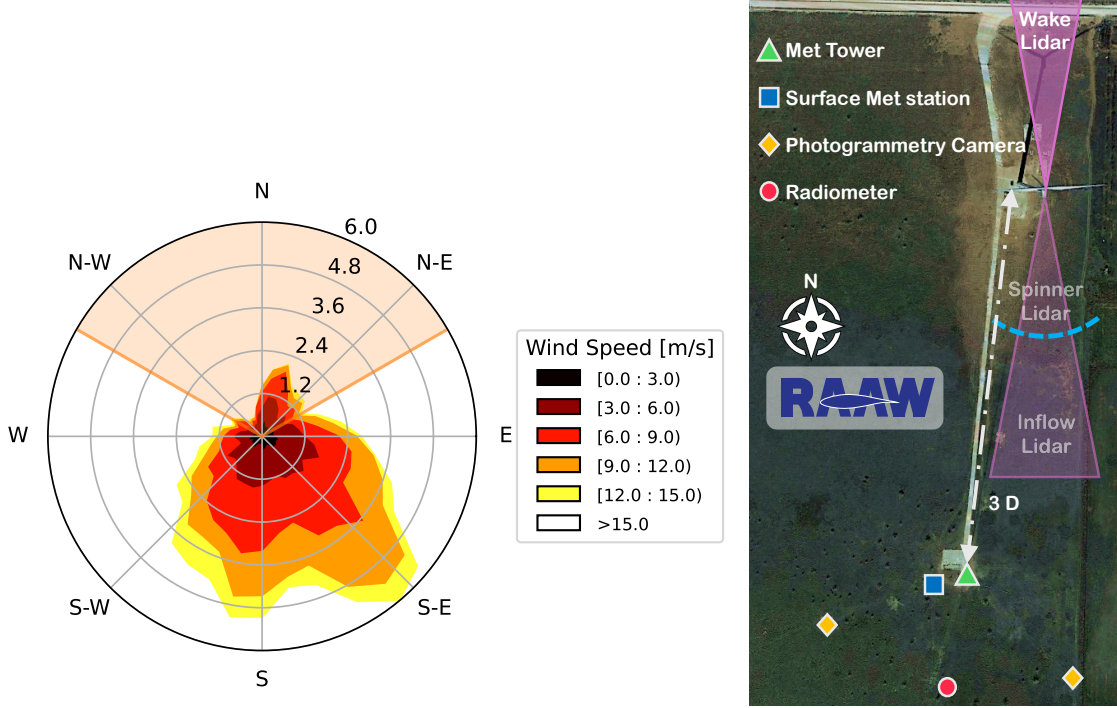


FIG. 3: Wind rose showing excluded wind direction sector in orange (left) and high-level schematic (right) of the RAAW project.

Lubbock, TX, that had a hub height of 120 m. In the AWAKEN project, the turbines are production-run GE2.8-127 turbines with a hub height of 88.5 m. In both projects, Halo Photonics Streamline XR+ Doppler scanning lidars were positioned on the aft sections of the nacelles of the turbines of both projects and were configured for a variety of measurement strategies, including a plan-position indicator (PPI) scan optimized to measure the fluctuating flow fields in the wake. Wake PPIs were designed using the LiSBOA tool²⁶, sweeping approximately 1° of azimuth per second converging a sector of approximately 20° centered downstream of the turbine. Lidar data were quality controlled using the FIeld EXperiments Tool Arsenal (FIEXTA) software⁴⁵ that includes the dynamic filtering methods outlined by Beck and Kühn⁴⁶. Data are temporally upsampled considering local advection velocities and the time delays between successive beams at similar azimuth angles,⁴⁷ producing line-of-sight velocity ‘snapshots’ at a temporal resolution of 1 second. Scanning lidar provides line-of-sight wind speed, v_{los} . The horizontal wind speed, u was retrieved as,

$$v_{los}(r) = u \cos \zeta \cos(\psi - \theta) + w \sin \zeta \quad (10)$$

where u is the horizontal wind speed, r represents the range along the laser beam, ζ is

the elevation angle, ψ is the azimuth angle, and θ , is the wind direction. For all lidar measurements used in the following analysis, the elevation angle was fixed at 0° , simplifying the relationship between v_{los} and u .

While the lidars are capable of making measurements up to 4 km from the wind turbine, the measurement quality is typically quite low, and wakes are often fully recovered far sooner. Additionally, the data collected in the AWAKEN project include only inflows from a southerly sector, and the downstream row of turbines is often visible in the scan data. To preclude observations of wakes of neighboring turbines from influencing wake center estimates, which would complicate the modal analysis undertaken below, scans are constrained to ranges where $2.5 \leq x/D \leq 10$ and azimuth angles $-20^\circ \leq \alpha \leq 20^\circ$. **check azimuth limits**

Each data set of continuous PPI scans lasted approximately 30 minute, with a full azimuthal sweep occurring approximately every 18 to 20 seconds. After spatiotemporal up-sampling following the procedure outlined in Beck and Kühn⁴⁷, each data case from the AWAKEN project contained approximately 1,650 snapshots (complete, infilled scans) at temporal resolution of 1 s. Data cases from the RAAW project are contained in 15 minute sections, leading to upsampled and infilled data sets of approximately 800 scans each at a temporal resolution of 1 s.

Measurement cases combining lidar data sets and inflow measurements were considered only after passing through the following quality control filters. Each filter is followed by a metric indicating the portion of the population of data that were disallowed and a description. Note that more than one filter condition was applicable to many of the data points.

Sensor Quality—6%: Measurement cases combining lidar data sets and inflow measurements were considered only after filtering out stuck, broken, or defective sensors. This ensures that all required measurements are reliable and eliminates cases where data are missing or sensors fail to report reasonable observations.

Wind Direction Sector—34%: Cases were filtered to include only wind directions within the appropriate sector. This is crucial because we rely on the characteristics of the inflow to contextualize wind turbine wake meandering. For the AWAKEN project, this step also ensures that only the wake dynamics from the single turbine of interest are present in the lidar scans.

Yaw Travel Limitation—51%: Only cases with a total yaw travel below 20° were considered. In both the RAAW and AWAKEN projects, the wind turbine operates at a nominal control point, allowing the rotor to yaw and optimally align with the incoming wind. Yaw activity is evident in lidar scans as abrupt changes or sweeping movements of the wakes within the scanned sectors. Very few lidar scan periods are completely free from yaw activity, hence the 20° travel limitation.

Power Production Threshold—18%: Cases were filtered to include only those where active power is greater than 150 kW. This criterion ensures that the turbine is operating within region 2 of the power curve and that a wake will be evident in the lidar data. The selection allows for describing wake meandering under a wide range of atmospheric conditions and inflow wind speeds.

Variability Filtering—10%: As a final quality control step, cases were excluded where the total variation exceeded the 90th percentile. Following the method outlined by Hamilton,⁴⁸ total variation is defined as the determinate of the covariance matrix relating inflow wind speed, wind direction, turbulence intensity, turbine power production, and nacelle position signals from the SCADA record. This step ensures that only statistically stationary conditions are considered, eliminating cases with drastically changing atmospheric conditions or significant turbine control actions during the lidar scan period.

A total of 65 cases were retained after applying each of the quality control filters enumerated above, including 54 cases from the AWAKEN project and 11 cases from the RAAW project. Figure 4 shows the distribution of the full range of cases with wake PPI scans. Cases that did not pass all of the quality control measures listed above are shown in gray. The final cases selected for analysis are highlighted in the figure according to the total yaw travel observed during the respective wake scanning period.

The integral time scale of atmospheric turbulence provides a measure of the temporal correlation of velocity fluctuations in the atmospheric boundary layer. The autocorrelation coefficient $\rho(\tau)$ is then defined as:

$$\rho(\tau) = \frac{\langle u'(t)u'(t + \tau) \rangle}{\sigma^2} \quad (11)$$

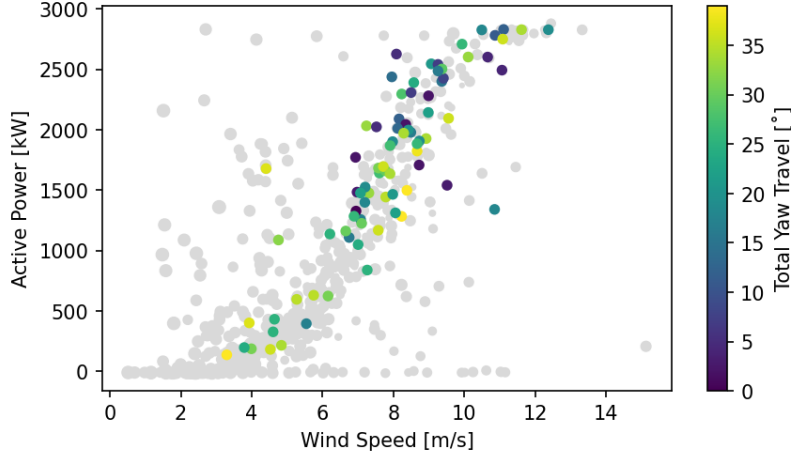


FIG. 4: Power curve from turbine H05 at the King Plains Wind Plant. All available cases are shown with gray points, cases selected for analysis in the current work are colored by the total yaw travel of the turbine during the measurement period.

where $u'(t)$ represents the turbulent velocity fluctuations, τ is the time lag, $\langle \cdot \rangle$ denotes the ensemble average, and σ^2 is the variance of the velocity fluctuations. The integral time scale T is calculated by integrating the autocorrelation function up to the point where it decays to a specified threshold:

$$T = \int_0^{\tau_c} \rho(\tau) d\tau \quad (12)$$

Here, τ_c represents the correlation decay limit, typically chosen as the point where $\rho(\tau) = 0.05$, rather than the first zero-crossing point.^{49,50} This integration captures the temporal coherence of turbulent fluctuations, providing insight into the characteristic time scales of atmospheric turbulence. In practice, this calculation was performed using 20-Hz wind speed measurements from a sonic anemometer positioned at hub height for the turbines in each project, with computations based on 30-minute averaging periods.

Figure 6 shows the distribution of inflow wind speed, U_∞ , integral time scale, T , and inflow Strouhal number, St_{in} for each of the 71 cases from the combined RAAW and AWAKEN experiments. The integral time scales, T , are taken as the characteristic period of the fluctuation in the velocity field and are used to define the characteristic frequency of turbulent fluctuations in the discussion of meandering frequencies.

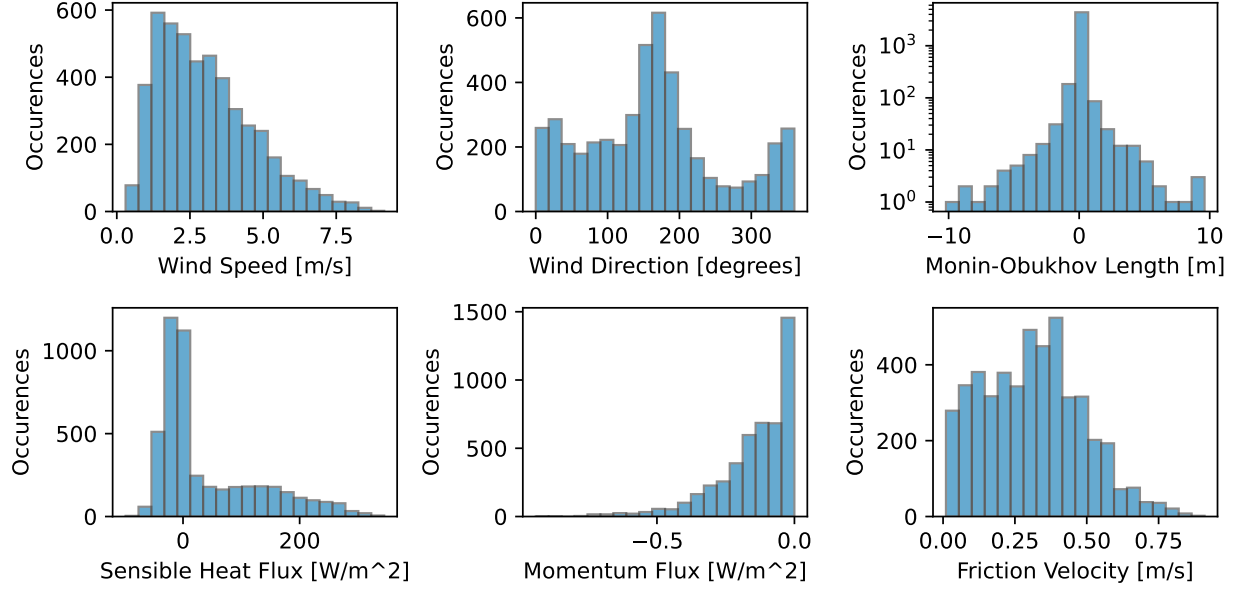


FIG. 5: Distributions of the average inflow conditions observed at the surface met stations at Site A1 throughout the AWAKEN and RAAW campaigns.

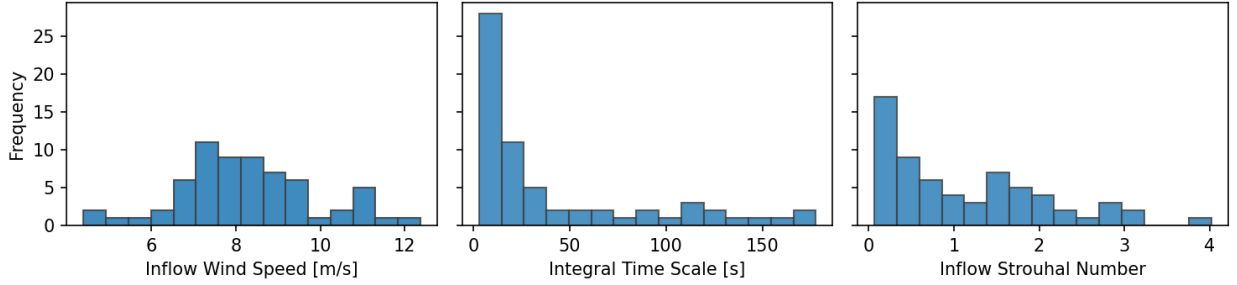


FIG. 6: Distributions of the average inflow wind speed, U_∞ , integral time scale, T , and Strouhal Number, St_{in} , at hub height for the subset of conditions corresponding to the 65 selected cases.

IV. RESULTS

Average scans shown for selected cases are shown in Fig. 7 and highlight differences in the peak momentum deficit and recovery rate of wakes in different atmospheric conditions. While all data sets include some yaw activity, the average wake is centered around $\alpha = 0$. Each of time-averaged velocity fields in Fig. 7 highlights characteristics of the wind turbine wake as well as the influence of the atmospheric forcing conditions and effects of the wind turbine operations and controls. The data quality control preferentially selected the most

statistically stationary cases from the full population of data. As a consequence of this selection, there is a bias toward stable atmospheric conditions in which the yaw travel was minimized during the lidar collection periods of both experiments.

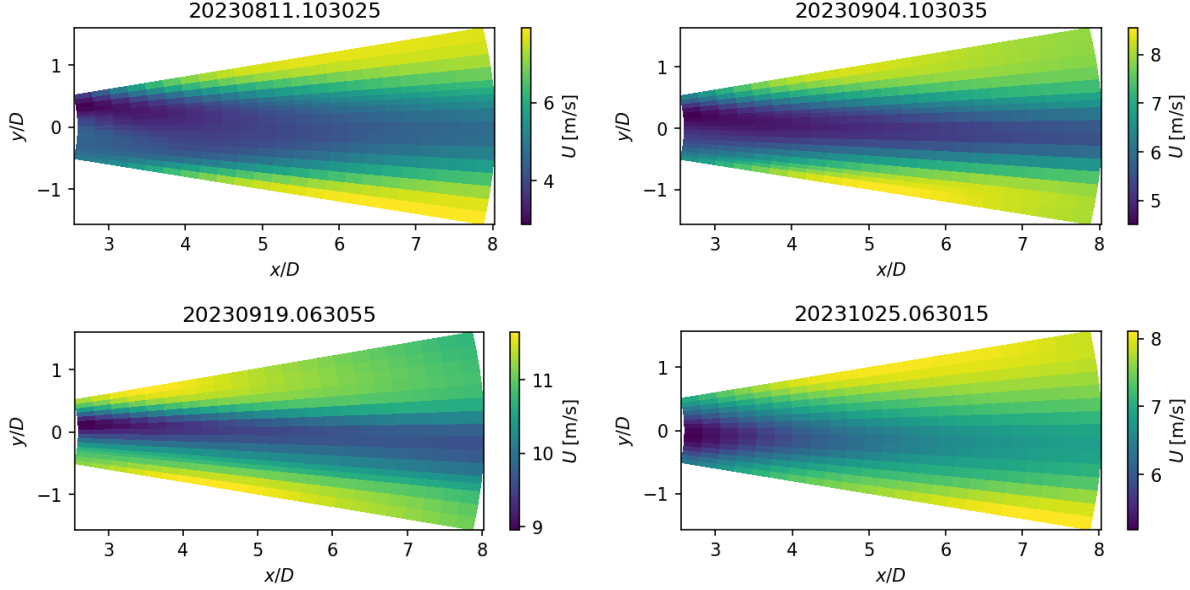


FIG. 7: Average wake velocity fields for 2023-08-11 10:30 (top left), 2023-09-04 10:30 (top right), 2023-09-19 06:30 (bottom left), 2023-10-25 06:30 (bottom right) showing a range of background velocities, momentum deficits and wake recovery rates.

Yaw activity is not directly evident in the time-averaged wakes. However, Fig. 8 shows a time-azimuth cross section of a single dataset at a range of approximately $x/D = 8$ (1 km), in which the wake is highlighted as a low wind speed region that persists in time across the figure. Meander in the azimuthal direction seen in the figure combines the effects of several factors. While the wake center does travel horizontally, there are also translations of the wake center introduced by the yaw activity of the rotor. Horizontal wake center locations, μ are estimated as the center location of the Gaussian function fit to the velocity deficit, as in Eq. (2). Yaw activity becomes especially evident in the time histories of detected wake centers, presented in Fig. 10. Yaw activity, illustrated by the dashed black line and related vertical axis on the right, are evident in the detected centers as sudden shifts in the wake center time histories. Because changes in the nacelle position impact the apparent wake centers uniformly at all ranges, a simple correction can be applied to the wake centers as, $\Delta\mu = r \sin \psi$, where r indicates the distance from the lidar (range) and ψ indicates the

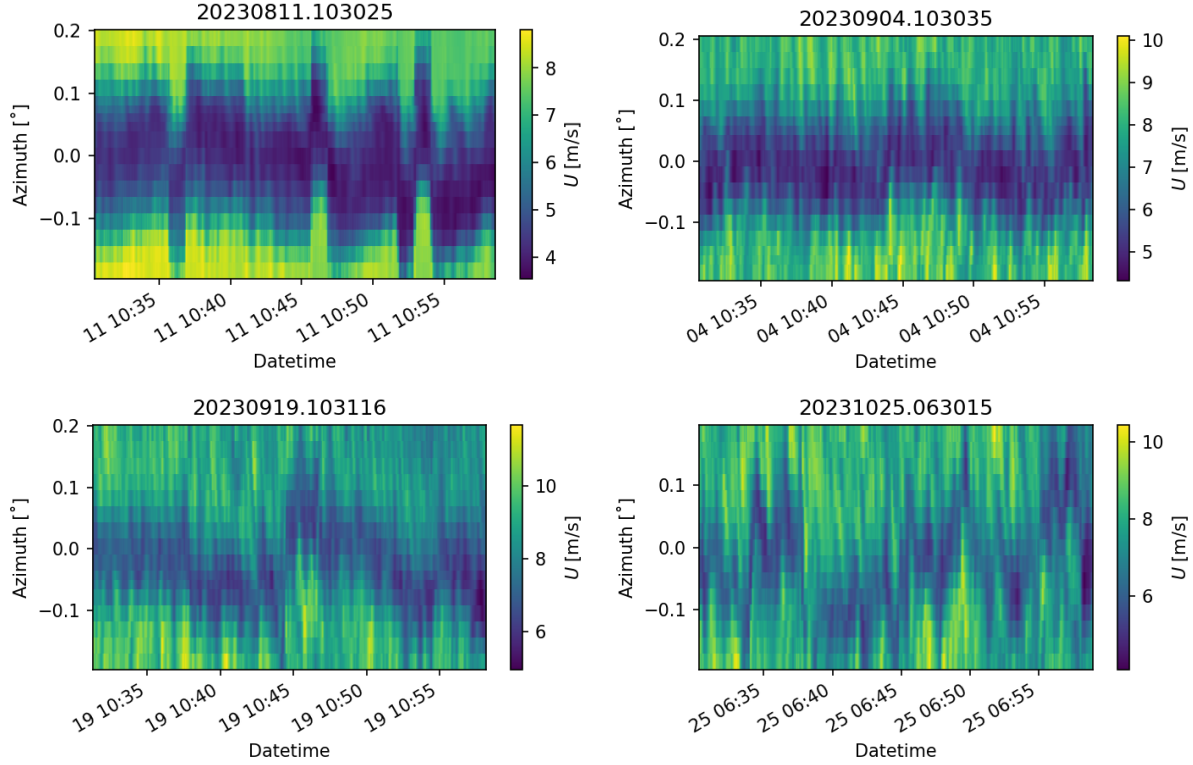


FIG. 8: Average wake velocity fields for 2023-08-11 10:30 (top left), 2023-09-04 10:30 (top right), 2023-09-19 06:30 (bottom left), 2023-10-25 06:30 (bottom right) showing a range of background velocities, momentum deficits and wake recovery rates.

nacelle orientation with respect to the position at the beginning of each data set. Correcting for changes in nacelle position throughout each dataset is important to correctly characterize the distributions of wake centers and their dynamics in the turbine frame of reference. It should be noted that turbines take yaw-correcting actions only after an integrator of error between the measured wind direction and the nacelle position exceeds a threshold. This means that in most cases there remains some influence of yaw misalignment on the detected wake centers even after correcting for yaw activity. Without a time history, it is not possible to accurately account for the yaw error. However, this sort of misalignment is part of the nominal operation of a turbine under normal control conditions, and is a possible trigger for wake meandering.

An example of the momentum deficit measured by the lidar is shown in Figure 11 in blue, along with local fits of the Gaussian function in black and detected wake centers as red points. Each of the subfigures indicate a good level of agreement between the momentum

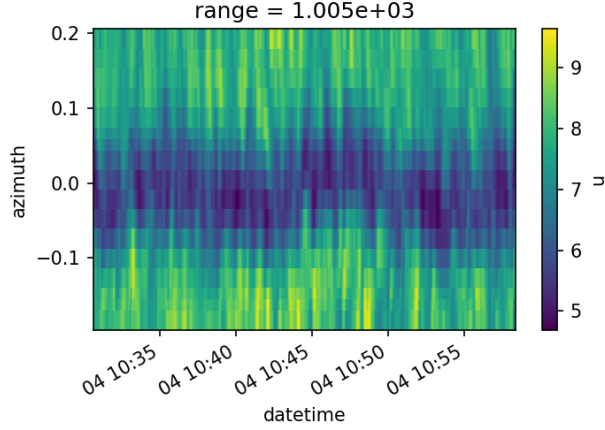


FIG. 9: Time-azimuth cross section of a line of sight velocity from a lidar scan at $8D$ downstream of the lidar. The wake is shown in dark blue and background velocity in green/yellow.

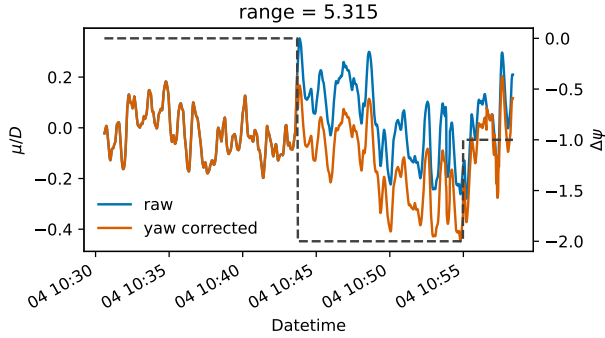


FIG. 10: Detected wake centers directly from lidar observations (blue) and after correcting for yaw activity of the turbine (orange). The dashed black line indicates nacelle position changes from the beginning of the time series and corresponds with the axis labels on the right.

deficit and fit function, building confidence in the Gaussian model of the wake. Alternative methods of detecting the wake centers have been explored in the literature,² including a momentum-centroid approach and local minimum velocity methods, but are not explored in the current work for brevity.

An alternative view of the detected wake centers is provided in Fig. 12, where μ/D is noted on the lidar scans as points colored by their respective R^2 values. In both examples, it is evident that the μ/D follows the blue momentum deficit in the wake quite closely. A bias of μ/D toward positive values of y/D in the near wake $x/D \lesssim 3$ is also visible in the

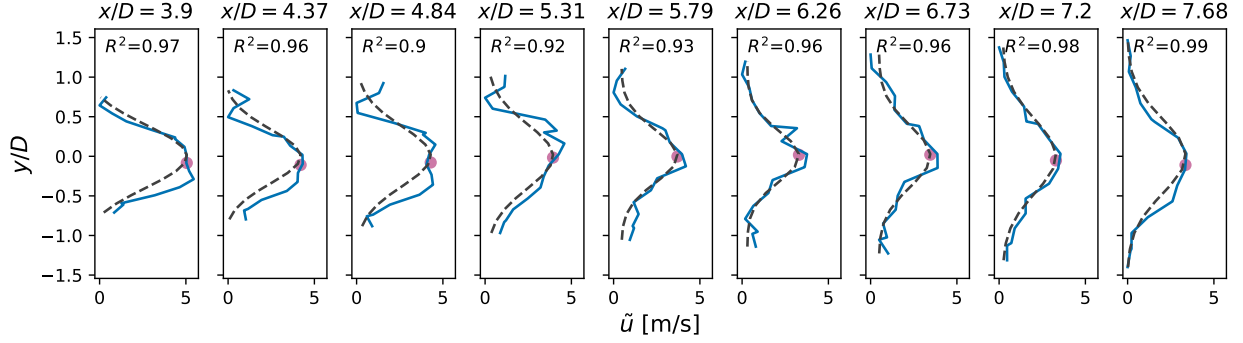


FIG. 11: Detected wake centers showing Gaussian fit against momentum deficit. Blue lines indicate momentum deficit estimated from lidar scans, dashed line shows the fit Gaussian profile, point marker is the detected wake center μ/D . **add R^2 value to each subplot**

scan taken at 2023-09-04 10:42:15 UTC (left). This bias, which persists throughout many of the scans, arises from the asymmetric double-Gaussian profile of the momentum deficit very near the turbine.^{51–53} Positive bias of μ/D in the near wake is not easily corrected, and leads to impacts later in the analysis, including power spectral densities an representation with the ROMs.

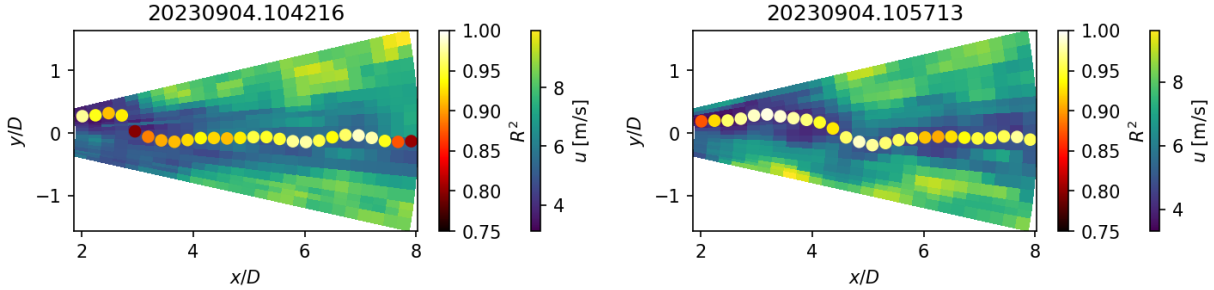


FIG. 12: Example lidar scans showing u overlaid with the fitted wake centers. Detected wake centers are colored by their respective coefficient of determination.

Wake meandering combines the downstream advection of μ by the mean flow and turbulent structures in the wake. Trajectories of μ throughout the measurement domain are shown in Fig. 13 (left) for several times in the data set. The trajectories all show the positive bias of μ in the near wake and then tend toward negative values of y as the range (downstream distance) increases. The meander is also evident in the trajectories as a local tendency back toward positive values of y . The amplitudes of these meandering structures also tend to increase as they advect downstream. Kernel density estimates of wake center

locations are shown in Fig. 13 (right), highlighting the tendency of the μ to spread out at greater streamwise distances.

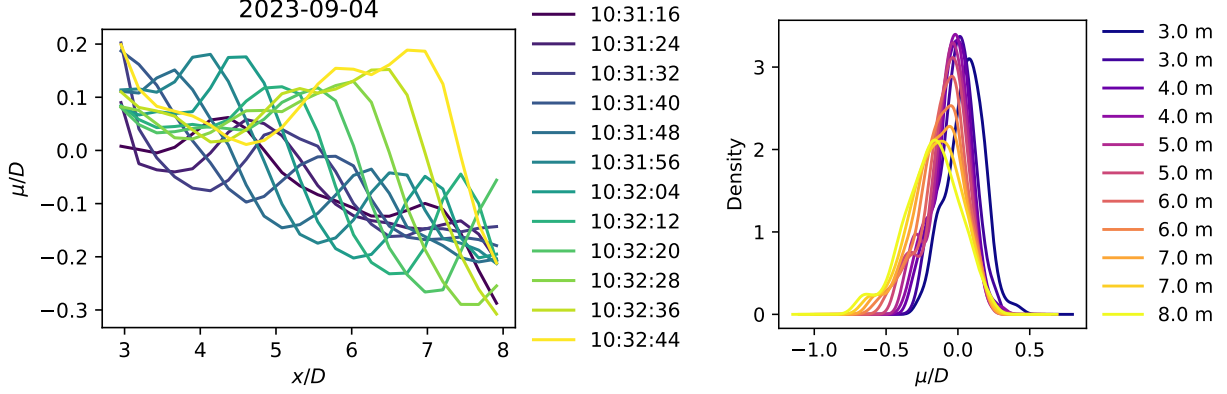


FIG. 13: Wake center trajectories for selected times (left) and kernel density estimates of wake centers by distance downstream of the lidar (right).

In addition to statistical descriptions of the wake center, we are also interested in dynamics of wake meandering, specifically relating the characteristic frequencies of wake meandering, quantified through power spectral densities (PSD, Eq. (13)) to inflow conditions and the POD modes, ϕ_i . PSDs are calculated using Welch's method as,

$$S_{\mu\mu}(f) = \frac{1}{L} \sum_{k=1}^K \frac{1}{N} \left| \sum_{n=1}^N \mu_k(n) e^{-i2\pi f n/N} \right|^2 \quad (13)$$

where, K is the number of segments, $L = 512$ samples is the number of points in the FFT, $N = 1,024$ is the length of each segment, $\mu_k(n)$ is the k -th segment of the time series. A uniform window function is applied to each segment in the calculation of the PSDs. The frequency range for all spectra in the current work have been normalized with the average inflow velocity recorded by the met towers, U_∞ , and the turbine's rotor diameter, D , as $f_{St} = f * D/U_\infty$.

The snapshot POD introduced in Section II B requires state space observations with consistent dimensions and coordinates, making it impractical to correct for yaw before decomposition. Remapping wake observations from lidar scans onto a corrected reference frame downstream of the turbine would introduce inconsistencies in the azimuthal coordinates or create empty radial slices, both of which would complicate the eigenvalue decomposition central to the POD analysis.

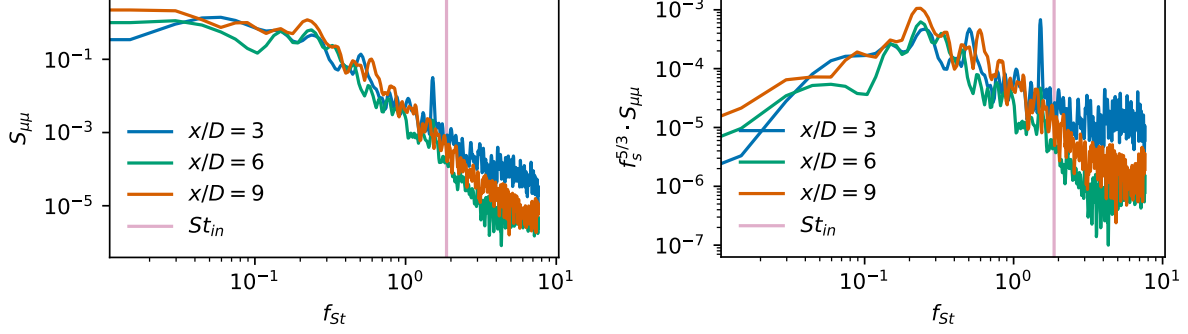


FIG. 14: PSDs (left) and premultiplied spectra (right) of detected wake centers from the lidar data at selected downstream distances. **fix vertical axis label for premult spectra**

Each of the 74 cases analyzed here has slightly varying azimuth coordinates due to the continuous scan operation of the lidar, which assigns nominal azimuth angles after a set time delta. Small variations in the number of scans per dataset arose from operational delays, data transmission, and status checks. Here, we focus on the low-rank POD modes, which account for the bulk of the turbulent kinetic energy (TKE) in the lidar scans. Although the energy distribution across POD modes differs for each dataset, the modes are ordered by descending energy, with the eigenvalue λ_i of each mode representing the energy distribution.

Only the first 14 POD modes, which contain the most coherent structures in the lidar scans, are used in flow field reconstructions for meandering analysis. The eigenvalues λ_i also quantify the TKE captured in each reduced-order model (ROM), providing a direct measure of reconstruction error. Missing energy in a ROM is bounded by the smallest mode eigenvalue included in the reconstruction. Figure 15 shows the distribution of TKE described by the POD modes across all cases. The energy of the zeroth mode ϕ_0 , which represents the mean flow field, is excluded, so the normalized sum represents only the turbulent kinetic energy. The black dashed line in the figure indicates the threshold at which 90% of the TKE is captured. For some cases, as few as 4 modes are required to reach this threshold, while other cases require more modes to reach the 90% TKE threshold. This threshold is met with a maximum of 14 modes, in all but two outlier cases.

Because we apply the POD to each lidar data set independently, the resulting analysis produces a unique basis of modes for each case. Figure 16 shows the POD modes associated with the case beginning at 2023-09-04 10:30 UTC. Although there are quantitative differ-

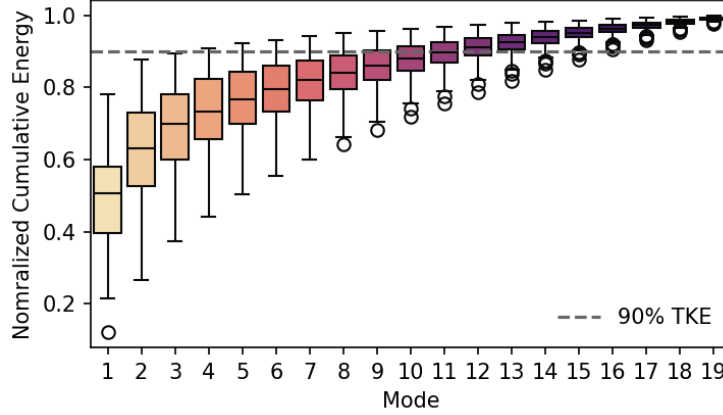


FIG. 15: The distribution of cumulative summations of eigenvalues associated with each case. Including

ences in the basis of modes for each case, qualitatively they are quite similar in structure and in order. Colorbars are included with each mode, for completeness, although it should be noted that it is only when combined with the respective mode coefficients to form a ROM that the units and quantitative values are meaningful.

Because the POD is applied independently to each lidar dataset, the analysis generates a unique set of modes for each case. Figure 16 displays the POD modes for a specific case starting at 2023-09-04 10:30 UTC. While there are quantitative differences in the mode bases across cases, their structure and order remain qualitatively similar. Each mode is shown with a colorbar for completeness, although the units and quantitative values become meaningful only when combined with the corresponding mode coefficients to form a ROM.

The zeroth mode, ϕ_0 , represents the mean flow and clearly depicts the characteristic momentum deficit region of a wind turbine wake. Modes ϕ_1 through ϕ_{14} describe coherent structures in the wake, ranked by their projection onto the lidar scans. The first mode, ϕ_1 , which accounts for approximately 20% to 8% of the TKE, captures large lateral shifts in the mean wake location and is most relevant for wake meandering. If we analogize the POD modes as Fourier modes, ϕ_2 , the second mode, describes a full oscillation in the streamwise (x) direction and a half oscillation in the transverse (y) direction. This mode, one order higher than ϕ_1 , represents the largest streamwise variations in the wake trajectory. Depending on the algebraic sign of its coefficient at any given time, ϕ_2 may describe a shift toward positive y near the turbine and toward negative y farther downstream, or the reverse.

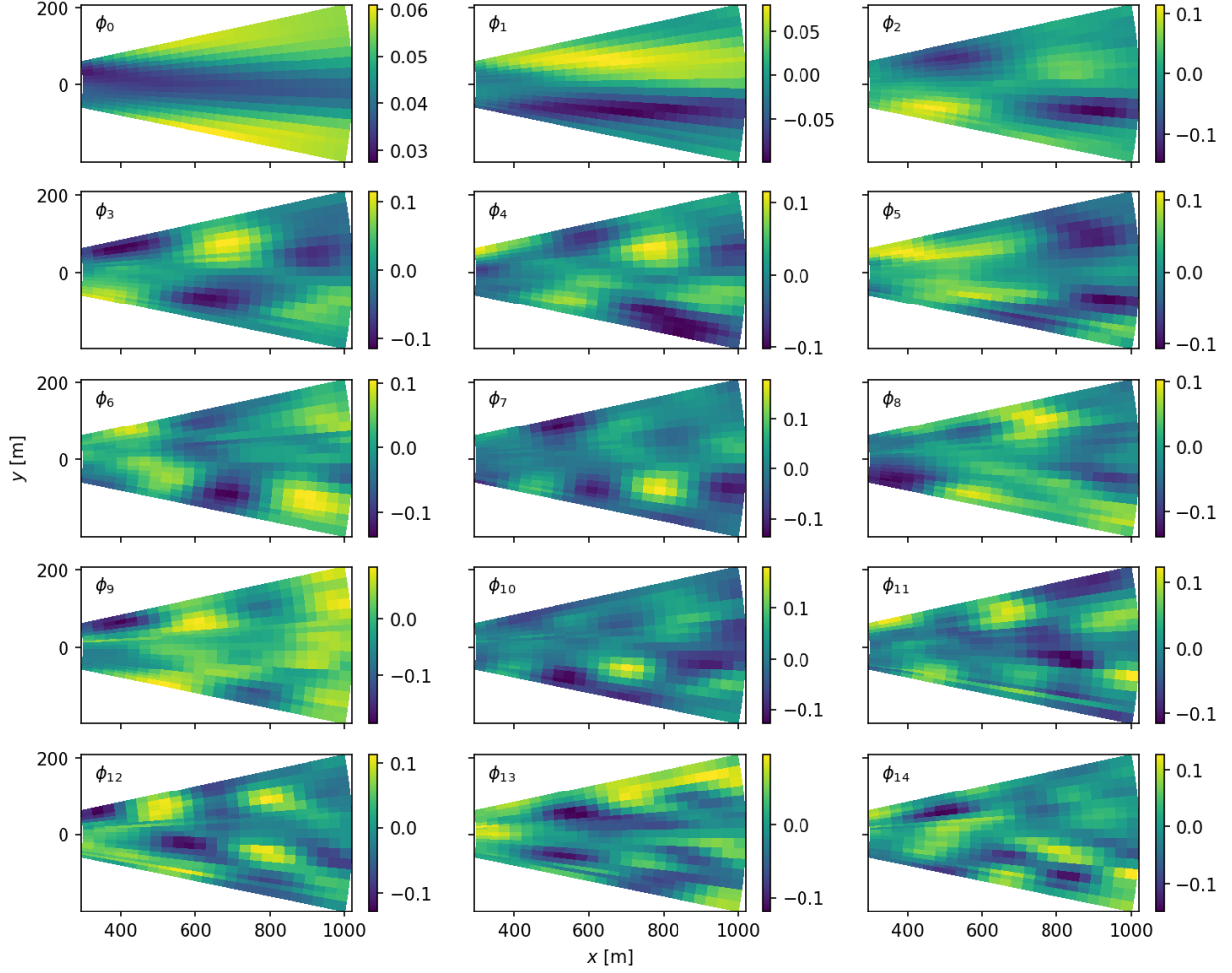


FIG. 16: POD modes from the example case on 2023-09-04 10:30 UTC.

The third mode, ϕ_3 , represents an additional half-period of oscillation, describing a period and a half in x across the lidar scan domain.

From the fourth mode onward, it becomes increasingly difficult to relate the modes to specific changes in wake structure. For instance, ϕ_4 is the first mode to display significant azimuthal asymmetry, complicating interpretation. In contrast, ϕ_5 exhibits azimuthal symmetry, a first for the modes presented in Fig. 16, but also displays a full oscillation along the x direction, possibly reflecting vertical wake meandering projected onto the plane of the lidar scans. A mode describing two periods of vertical meandering is observed in ϕ_9 , further illustrating the complexity of higher-order modes.

In reconstructing the flow field following Eq. (9), between 3 and 14 POD modes were used to describe the turbulence, in addition to the mean flow described in ϕ_0 . For each case, all

16,278 possible mode combinations were tested, producing a large sample of reconstructed wakes, each described by a unique subset of the modes from ϕ_1 to ϕ_{14} . Figure 17 illustrates the mode combinations used in each ROM, with ROMs defined by only four modes (the mean flow ϕ_0 and three turbulent modes between ϕ_1 and ϕ_{14}) displayed on the left, and those incorporating more modes shown on the right. For each combination, the wake profiles were fitted to the Gaussian function in Eq. (2), and a low-order estimate of the lateral wake center, $\hat{\mu}$, was extracted.

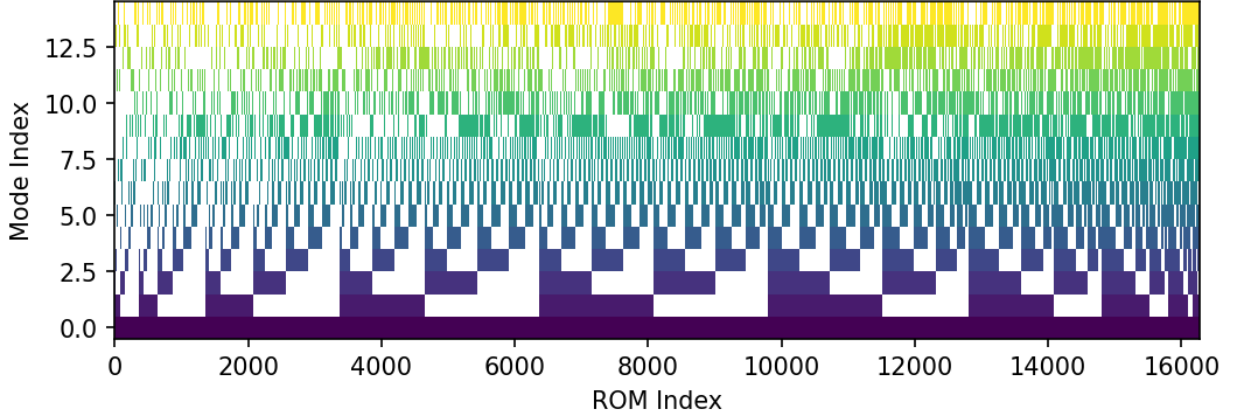


FIG. 17: Combination tree indicating which modes are contained in any given ROM. Modes are indicated by the colored regions, and their absence is shown in white. **change cbar label**

When reconstructing the velocity field using POD modes, our objective is to match the power spectral density (PSD) of wake center locations from the reconstructed flow field, $S_{\hat{\mu}\hat{\mu}}$, as closely as possible to that of the original field, $S_{\mu\mu}$. Mode combinations that accurately reproduce wake meandering for each case (i.e., where $S_{\mu\mu} \approx S_{\hat{\mu}\hat{\mu}}$) are considered representative of wake meandering in the low-order velocity field representation. Coherence, defined in Eq. (14), quantifies the similarity between wake center trajectories in the lidar scans and the ROMs.

$$C_{\mu\hat{\mu}}(f, x) = \frac{|S_{\mu\hat{\mu}r}(f, x)|^2}{S_{\mu\mu}(f, x)S_{\hat{\mu}\hat{\mu}}(f, x)} \quad (14)$$

In this context, coherence $C_{\mu\hat{\mu}}(f, x)$ measures the similarity of the wake center dynamics at frequency f , where $C_{\mu\hat{\mu}}(f, x) = 1$ indicates a perfect match between the PSDs of the wake centers, and $C_{\mu\hat{\mu}}(f, x) = 0$ indicates no match.

As a general trend, the PSDs of the wake centers show better agreement at lower fre-

quencies ($f_s < St_{in}$ Hz), which are more accurately captured by the POD modes used in the ROMs. Coherence decreases at higher frequencies, but because wake meandering is predominantly a low-frequency phenomenon, this reduction in coherence at higher f_s is acceptable for the purposes of this study. To illustrate this, Fig. 18 shows the average coherence over the full frequency range for several cases, which demonstrate good agreement between $S_{\mu\mu}$ and $S_{\hat{\mu}\hat{\mu}}$, particularly for $x/D > 4$. The highlighted regions in Fig. 18 indicate the region where $f_s < St_{in}$, that is where the reduced frequency is less than twice the inflow Strouhal number. Considering only part of the total frequency range of $C_{\mu\hat{\mu}}$ gives a more accurate representation of the quality of any particular ROM to reproduce wake meandering in particular—the high frequency range of the spectra is not relevant to wake meandering as the physics of interest. The trends for both ROMs in Fig. 18 were evaluated by calculating the average coherence, $\overline{C_{\mu\hat{\mu}}}$, over the respective frequency range.

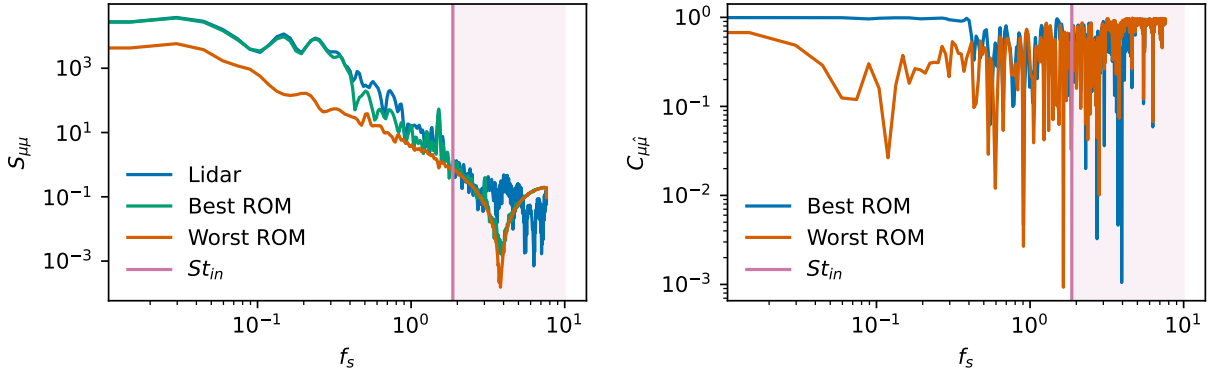


FIG. 18: Spectra (left) and coherence (right) for the ROMs that compare best (green) and worst (orange) to the PSD of μ from the lidar scans.

A more complete picture of ROM quality is provided in Fig. 19, which shows the coherence averaged over the reduced frequency, $\overline{C_{\mu\hat{\mu}}}$. The top subfigure shows $\overline{C_{\mu\hat{\mu}}}$ considering the full frequency range, and suggests that the average coherence is around 0.3 for most cases. In contrast, the bottom subfigure shows $\overline{C_{\mu\hat{\mu}}}$ considering only $f_s \leq St_{in}$, and shows much more consistent results for blocks of ROMs. Recalling the distribution of modes considered in each ROM presented in the combination tree shown in Fig. 17, a correlation between $\overline{C_{\mu\hat{\mu}}}$ and the presence of certain modes becomes clear.

The information contained in Fig. 19 also indicates dependence of $\overline{C_{\mu\hat{\mu}}}$ on the downstream distance away from the lidar, as shown in the vertical axes. Aggregating this information

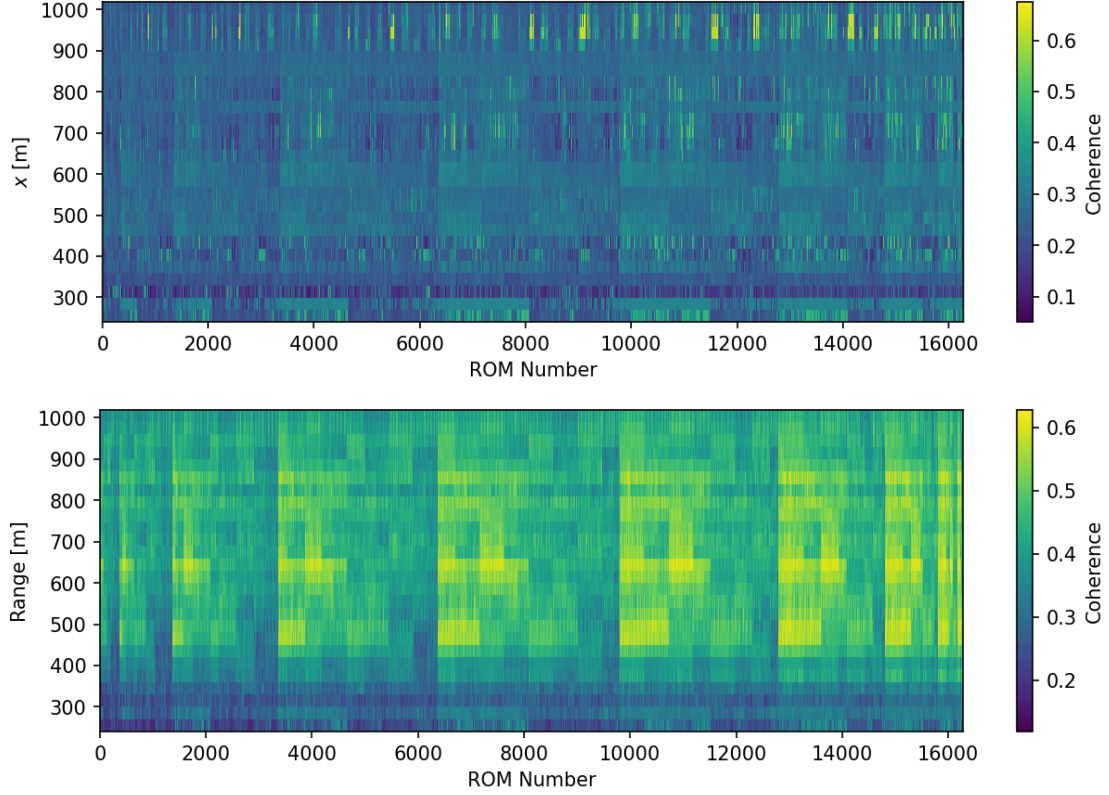


FIG. 19: Frequency-average coherence between detected wake centers from lidar scans and ROMs considering the full measured frequency range (top) and only frequencies $f_s \leq St_{in}$ Hz (bottom).

into a single scalar metric for each ROM by averaging along the range coordinate, $\varepsilon = \langle \overline{C}_{\mu\hat{\mu}} \rangle$, we are able to describe the distribution of ROMs in terms of a quantitative metric for fit quality. Figure 20 (left) shows ε arranged by the index of the ROM, spanning the 16,278 mode combinations. The data in Fig. 20 represent the the same data shown in Fig. 19, averaged over the streamwise coordinate (vertical plot axis). Aggregated a different way, Fig. 20 (right) shows the distribution of the ROM quality according to the number of modes used in the definition of each ROM. While the overall trend suggests that including more modes in a ROM generally increases the quality of the representation of wake meandering, it is interesting to note that there are plenty of exceptions to this trend. The distributions in Fig. 20 (right) indicate several ROMs with only four modes outperform some of the ROMs with 14 modes.

To evaluate the influence of individual POD modes on wake meandering, we perform a linear regression analysis using the full set of 16,287 mode combinations for each case. Given

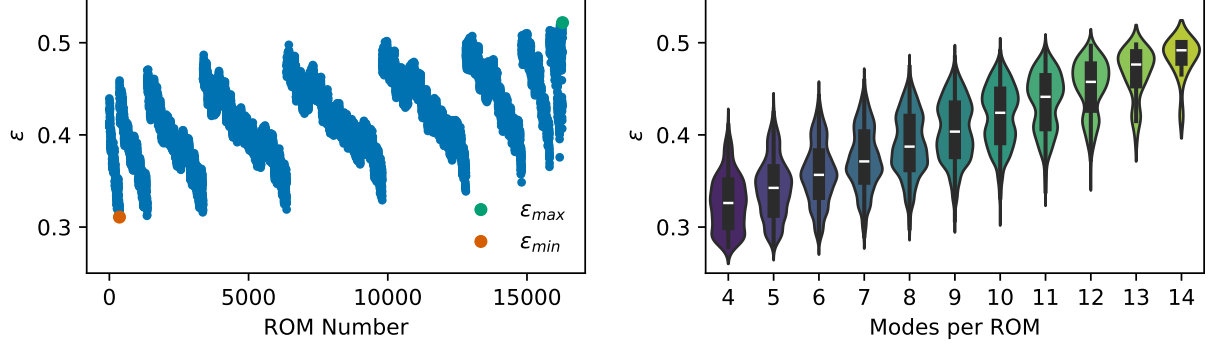


FIG. 20: Frequency and range-averaged coherence between detected wake centers from lidar scans and ROMs considering $f_s \leq St_{in}$ Hz (left). Distribution of ε by the number of modes contained in each ROM (right). **make vertical scales identical**

that each low-dimensional flow field representation is a linear combination of POD modes and their corresponding coefficients, the reconstructed flow field $\hat{u}$ is inherently linearly dependent on the included modes. The sensitivity of the wake center location to the inclusion or exclusion of specific modes was examined through a linear regression of modes (as dummy variables) against the average coherence between μ and $\hat{\mu}$, $C_{\mu\hat{\mu}}(f, x)$.

The sensitivity of the wake center coherence to POD modes was investigated using an Analysis of Variance (ANOVA⁵⁴) approach. We develop a statistical model to decompose the observed coherence metric into contributions from individual modes and their interactions. The model takes the form of a linear decomposition, where the average coherence ε as a quality metric for each ROM is expressed as a function of mode presence, accounting for both individual mode effects and two-way interactions:

$$\varepsilon = \bar{\varepsilon} + \sum_{i=1}^n \beta_i q_i + \sum_{i < j}^n \gamma_{ij} q_i q_j + \xi \quad (15)$$

where, $\bar{\varepsilon}$ represents the global mean coherence for each case, β_i are the main effects of individual modes, and γ_{ij} capture the interaction effects between modes. The binary variables q_i indicate the presence (1) or absence (0) of each mode, with a constraint that at least three modes must be included. The term ϵ represents the residual error.

We impose statistical constraints assuming that both the main effects β_i and interaction terms γ_{ij} follow a normal distribution with zero mean, and that the residual error ϵ follows a normal distribution. The objective is to estimate the coefficients β_i and γ_{ij} that minimize the residual error while providing insight into how different modes and their combinations

influence the wake center coherence. This approach allows for a comprehensive sensitivity analysis that goes beyond simple linear contributions, capturing the complex, potentially non-linear interactions between POD modes in representing the wake center dynamics.

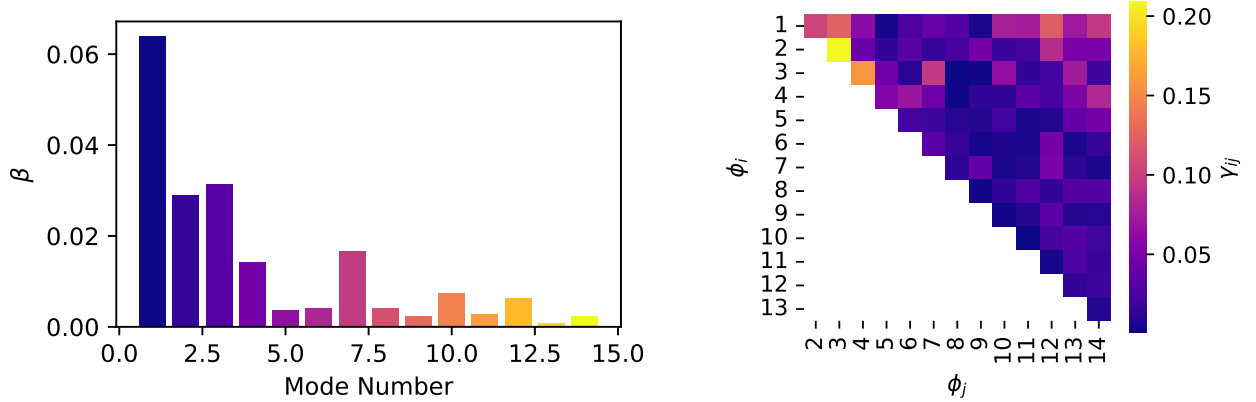


FIG. 21

V. CONCLUSIONS

ACKNOWLEDGEMENTS

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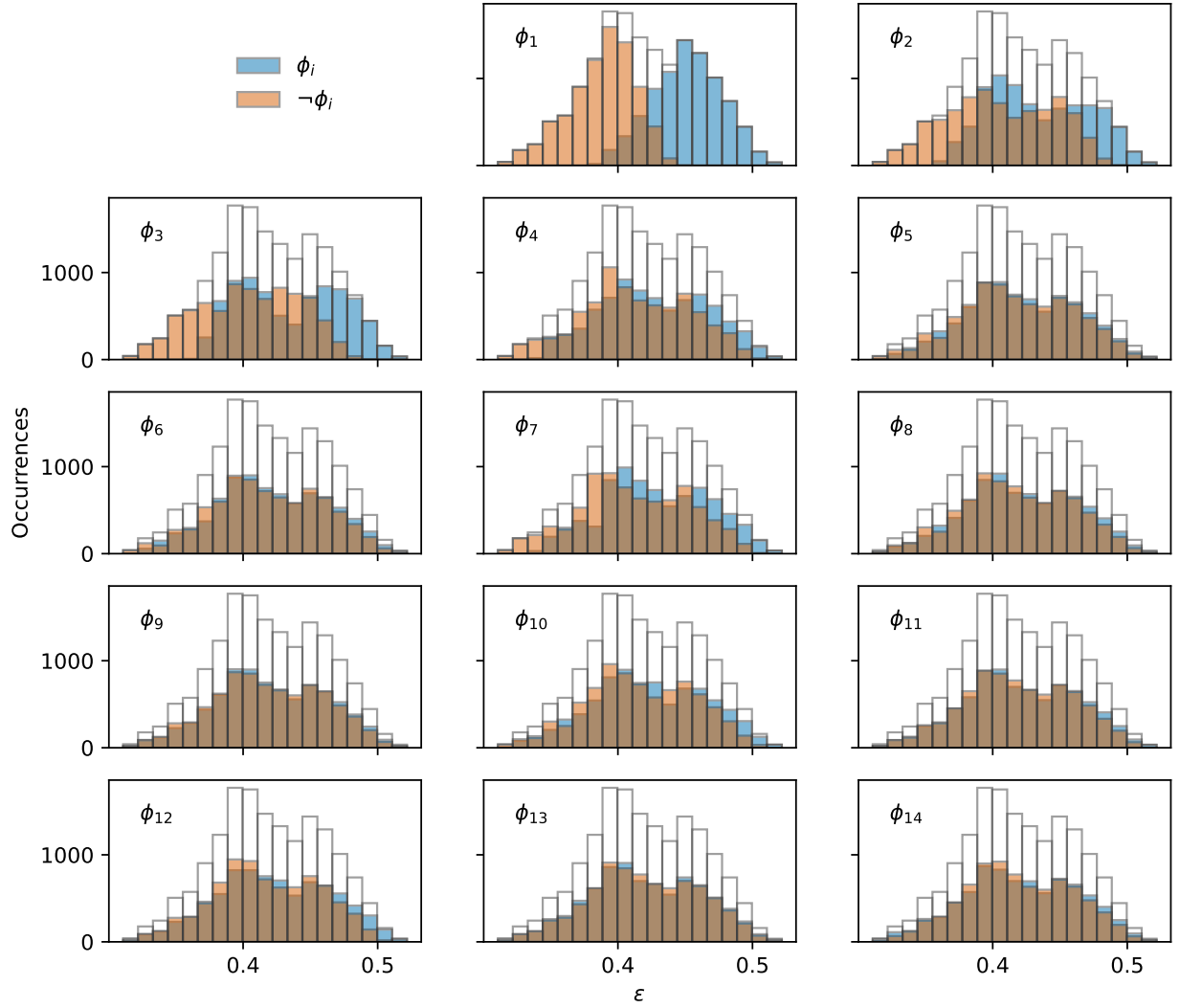


FIG. 22

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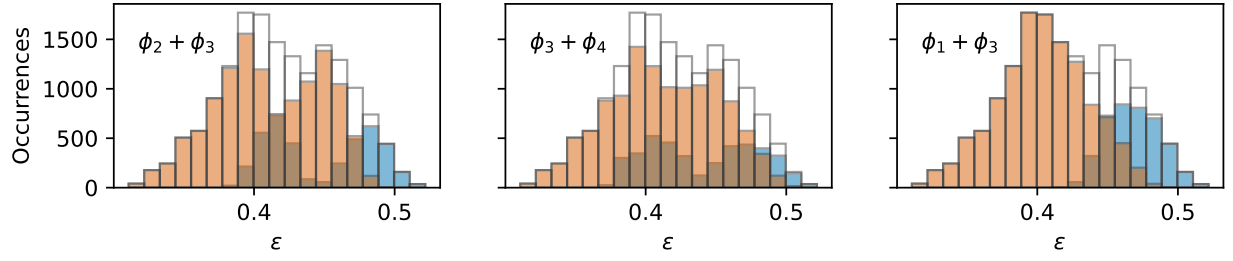


FIG. 23

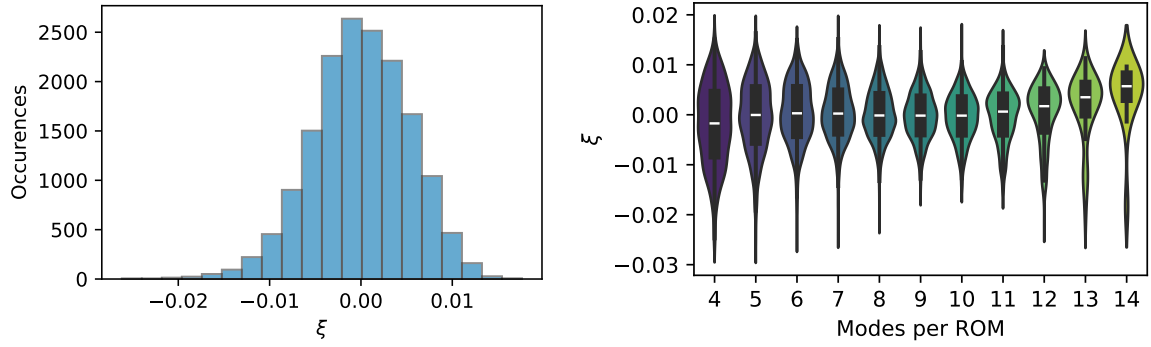


FIG. 24

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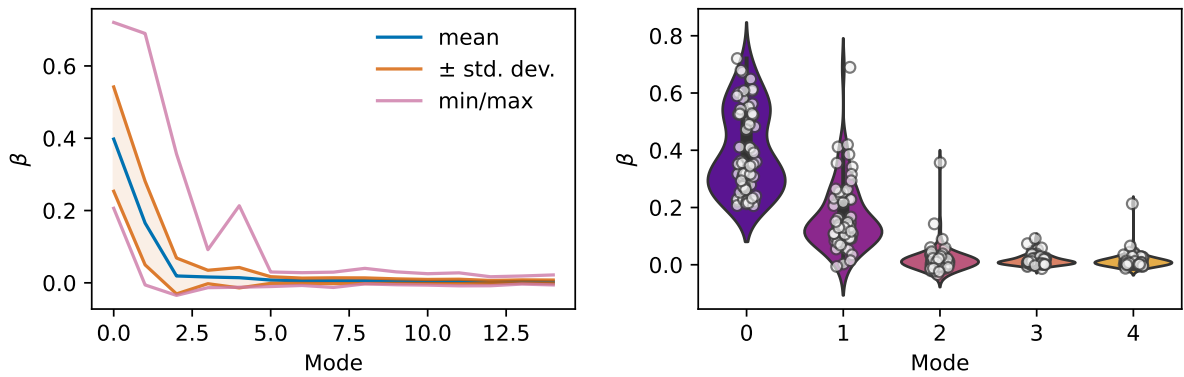


FIG. 25

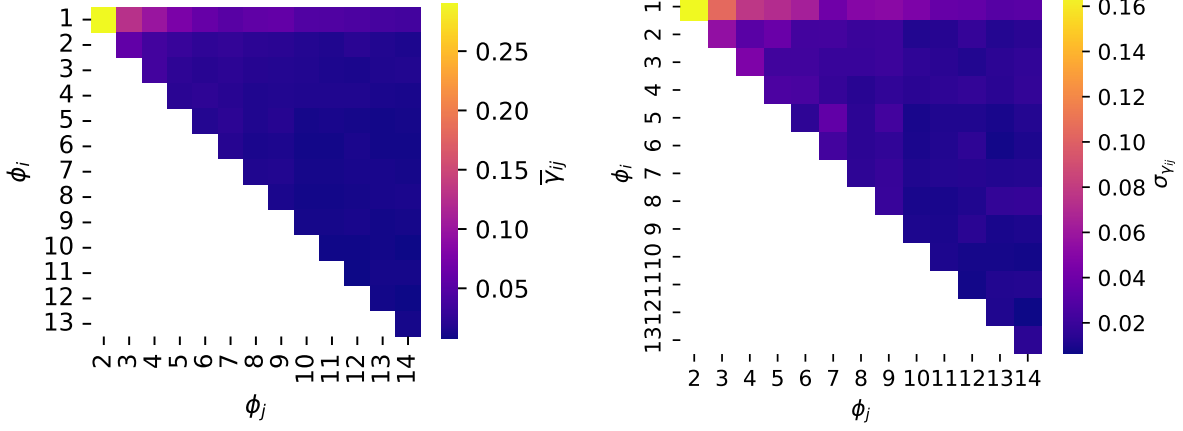


FIG. 26

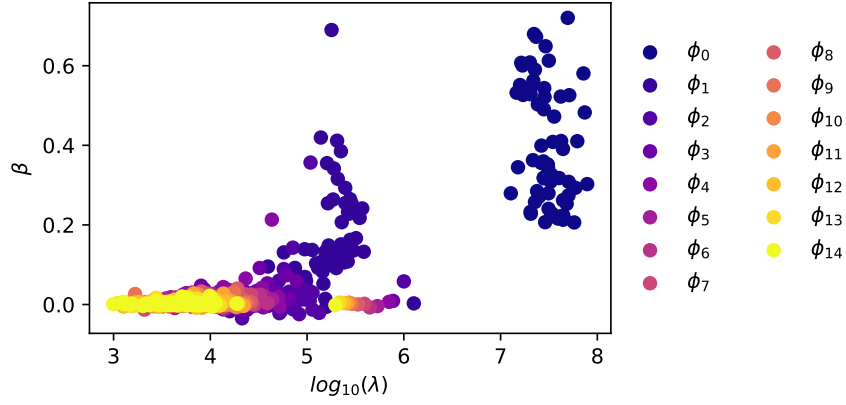


FIG. 27

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